

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data into Delta Tables

- Title: Load Customer and Order Data into Delta Tables
- Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.
- Preconditions:

CSV files are available at the specified locations:

``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and
``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``.

- The Delta tables for customer and order data do not exist or are empty.
- Main Flow / Functional Steps:
- Read customer CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` into a Delta table named "customer".
- Read order CSV data from ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` into a Delta table named "order".

The schema for the "customer" table is: CustId, Name, EmailId, Region.

The schema for the "order" table is: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-DLT-002: Transform Order Data by Adding TotalAmount Column

- Title: Transform Order Data by Adding TotalAmount Column
- Description: The system must add a new column "TotalAmount" to the order table, which is the product of "PricePerUnit" and "Qty" columns.
- Preconditions:
- The "order" Delta table exists and is populated with data.
- Main Flow / Functional Steps:
- Add a new column "TotalAmount" to the "order" table.
- Calculate the value of "TotalAmount" by multiplying "PricePerUnit" and "Qty" columns.

FR-DLT-003: Cleanse Customer and Order Data

- Title: Cleanse Customer and Order Data
- Description: The system must remove null and duplicate records from both customer and order tables.
- Preconditions:
 - The "customer" and "order" Delta tables exist and are populated with data.
- Main Flow / Functional Steps:
 - Remove rows with null values from both "customer" and "order" tables.
 - Remove duplicate records from both "customer" and "order" tables.

FR-DLT-004: Create ordersummary Table and Load Joined Data

- Title: Create ordersummary Table and Load Joined Data
- Description: The system must create an "ordersummary" table if it does not exist and load joined data from customer and order tables into it using SCD type 2.
- Preconditions:
 - The "customer" and "order" Delta tables exist and are populated with cleansed data.
 - The "ordersummary" table does not exist or is empty.
- Main Flow / Functional Steps:
 - Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdhc_wizard".
 - Join the "customer" and "order" tables on the "CustId" field.
 - Load the joined data into the "ordersummary" table using SCD type 2.

FR-DLT-005: Update ordersummary Table on Customer Data Change

- Title: Update ordersummary Table on Customer Data Change
- Description: The system must update the "ordersummary" table whenever there is a change in the customer table.
- Preconditions:
 - The "customer" and "ordersummary" tables exist and are populated with data.
- Main Flow / Functional Steps:
 - Detect changes in the "customer" table.
 - Update the "ordersummary" table by making old records inactive and new records active.
 - Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-006: Create customeraggregatespend Table and Load Aggregated Data

- Title: Create customeraggregatespend Table and Load Aggregated Data
- Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from the "ordersummary" table into it.
- Preconditions:
 - The "ordersummary" table exists and is populated with data.
 - The "customeraggregatespend" table does not exist or is empty.
- Main Flow / Functional Steps:
 - Create the "customeraggregatespend" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
 - Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" columns.
 - Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-007: Implement Data Pipelines using Delta Live Tables

- Title: Implement Data Pipelines using Delta Live Tables
- Description: The system must implement the data pipelines for loading and transforming data using Delta Live Tables.
- Preconditions:
 - All previous requirements are implemented and functioning correctly.
- Main Flow / Functional Steps:
 - Implement the data pipelines using Delta Live Tables to load and transform data as per the previous requirements.